Algebraic Limits: Factoring and Rationalizing

AP Calculus AB/BC

Each limit below is indeterminate of the form $\frac{0}{0}$ on direct substitution. Rewrite first, then substitute. Show the algebra that removes the common factor.

1. Evaluate $\displaystyle\lim_{x\to 3} \frac{x^2 - 9}{x - 3}$.

The table below shows the function approaching $x = 3$. Use it to predict the limit, then confirm it algebraically.

x	2.9	2.99	3	3.01	3.1
$f(x)$	5.9	5.99	undefined	6.01	6.1

Answer: ________________

2. Evaluate $\displaystyle\lim_{x\to 2} \frac{x^2 - 5x + 6}{x - 2}$.

Answer: ________________

3. Evaluate $\displaystyle\lim_{x\to 2} \frac{x^3 - 8}{x - 2}$. (Hint: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.)

Answer: ________________

4. Evaluate $\lim_{x\to 0} \dfrac{\sqrt{x+4}-2}{x}$ by multiplying by the conjugate.

Answer: ________________

5. Evaluate $\lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9}$.

Answer: ________________

6. Evaluate $\lim_{x\to 1} \dfrac{2-\sqrt{x+3}}{x-1}$. Watch the sign: the radical is being subtracted *from* a constant.

Answer: ________________

7. Let

$$f(x) = \begin{cases} \dfrac{x^2 - 5x - 14}{x - 7}, & x \neq 7 \\ c, & x = 7 \end{cases}$$

Find the value of c that makes f continuous at $x = 7$.

Answer: ________________

8. In problem 1 you cancelled the factor $(x - 3)$ from $\dfrac{x^2 - 9}{x - 3}$ to get $x + 3$. Explain, in a sentence or two, why that cancellation is legitimate when computing the limit, even though $\dfrac{x^2 - 9}{x - 3}$ and $x + 3$ are *not* the same function at $x = 3$.